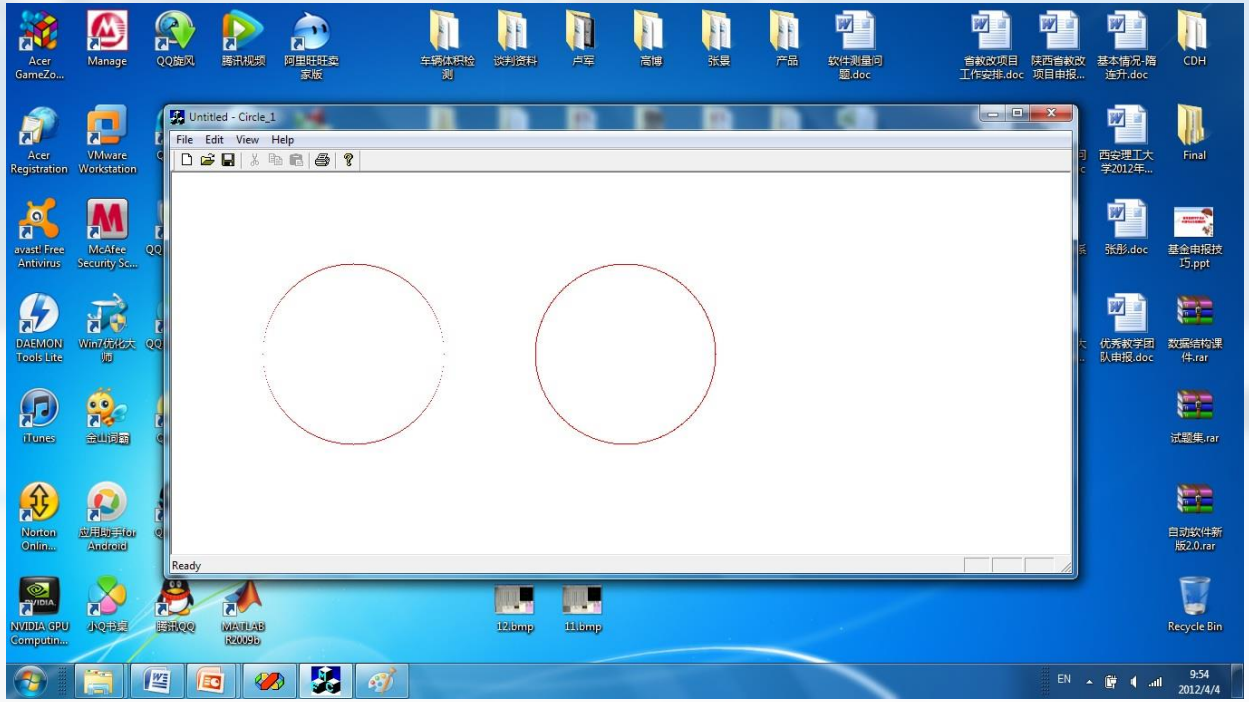
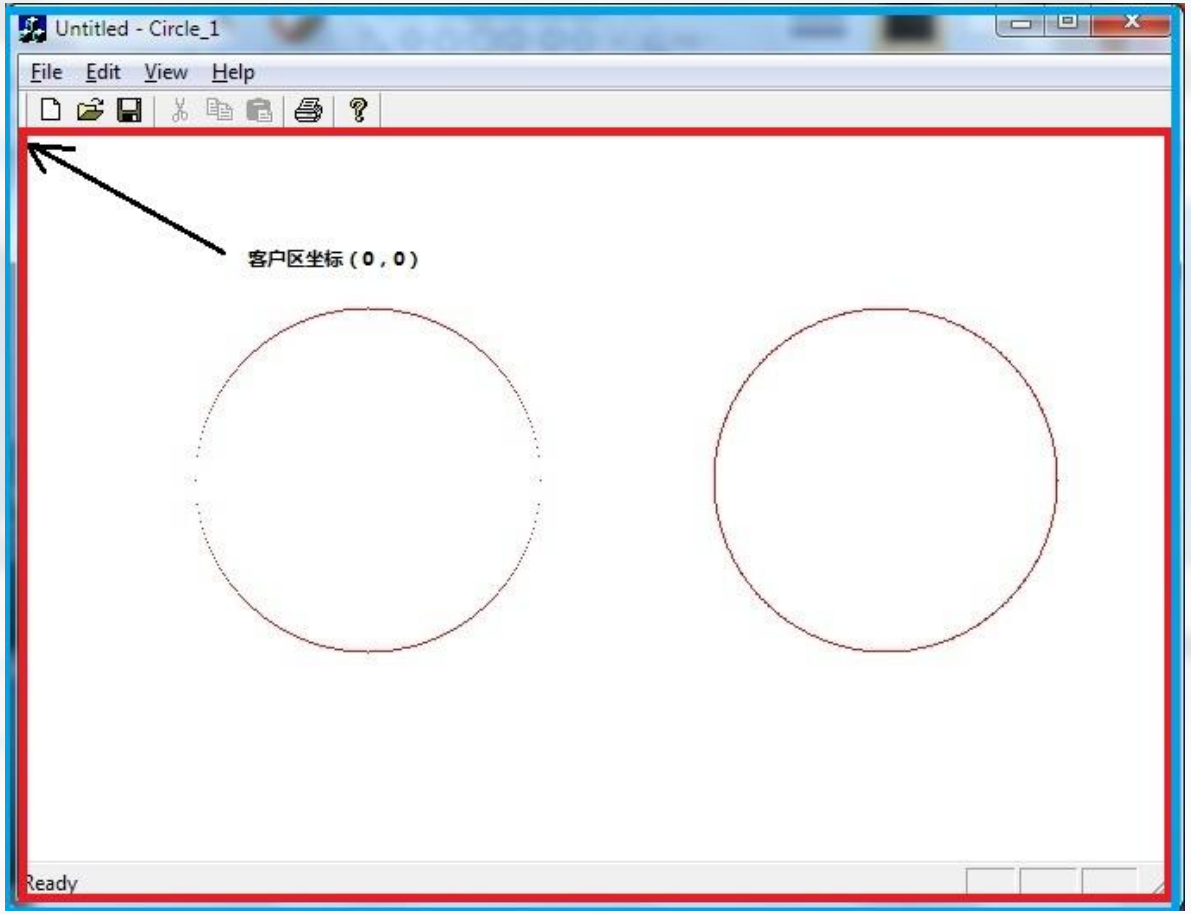
交互技术

Windows的应用程序运行时，当一个窗口建立后，显示屏幕被划分成三个区域：屏幕区、窗口区、客户区。



交互技术



交互技术

MFC提供的绘图类通常分成两大部分：一部分是GDI类，另一部分是CDC类。

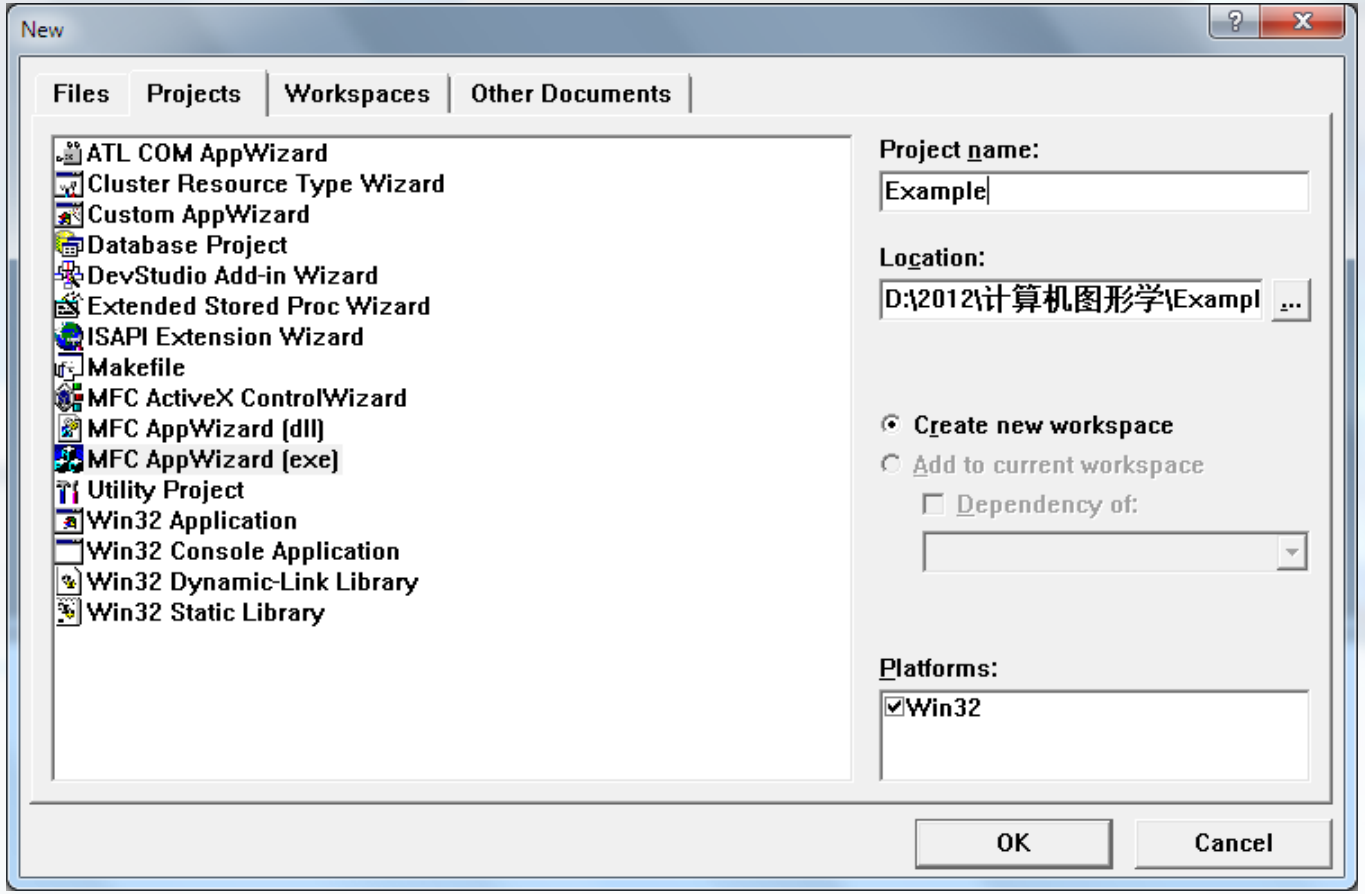
图形设备接口类（GDI类）：CBitmap、CBrush、CFont、CPalette、CPen、CRgn

设备描述表类（CDC类）：CClientDC、CMetaFileDC、CPaintDC、CWindowDC

交互技术

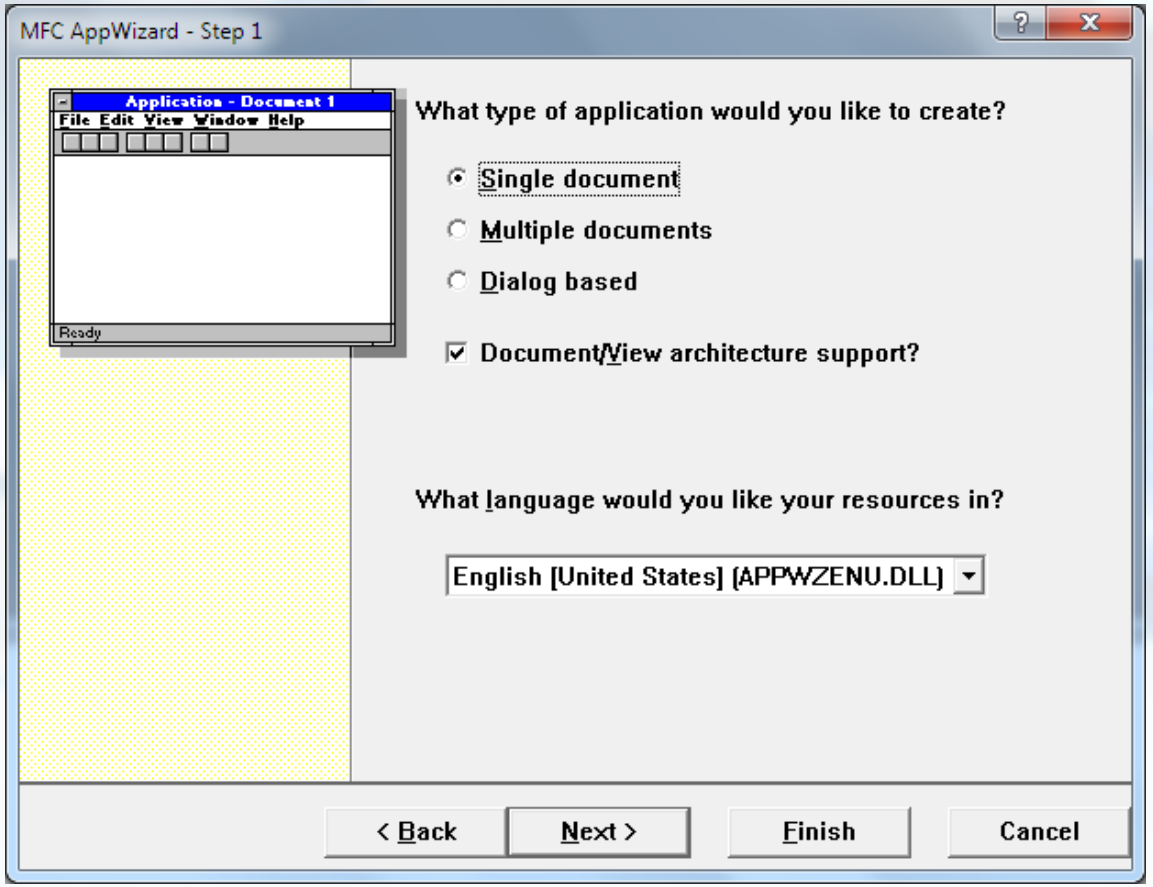
基于MFC编写图形系统：Windows程序的运行是基于消息的。

基本过程如下：



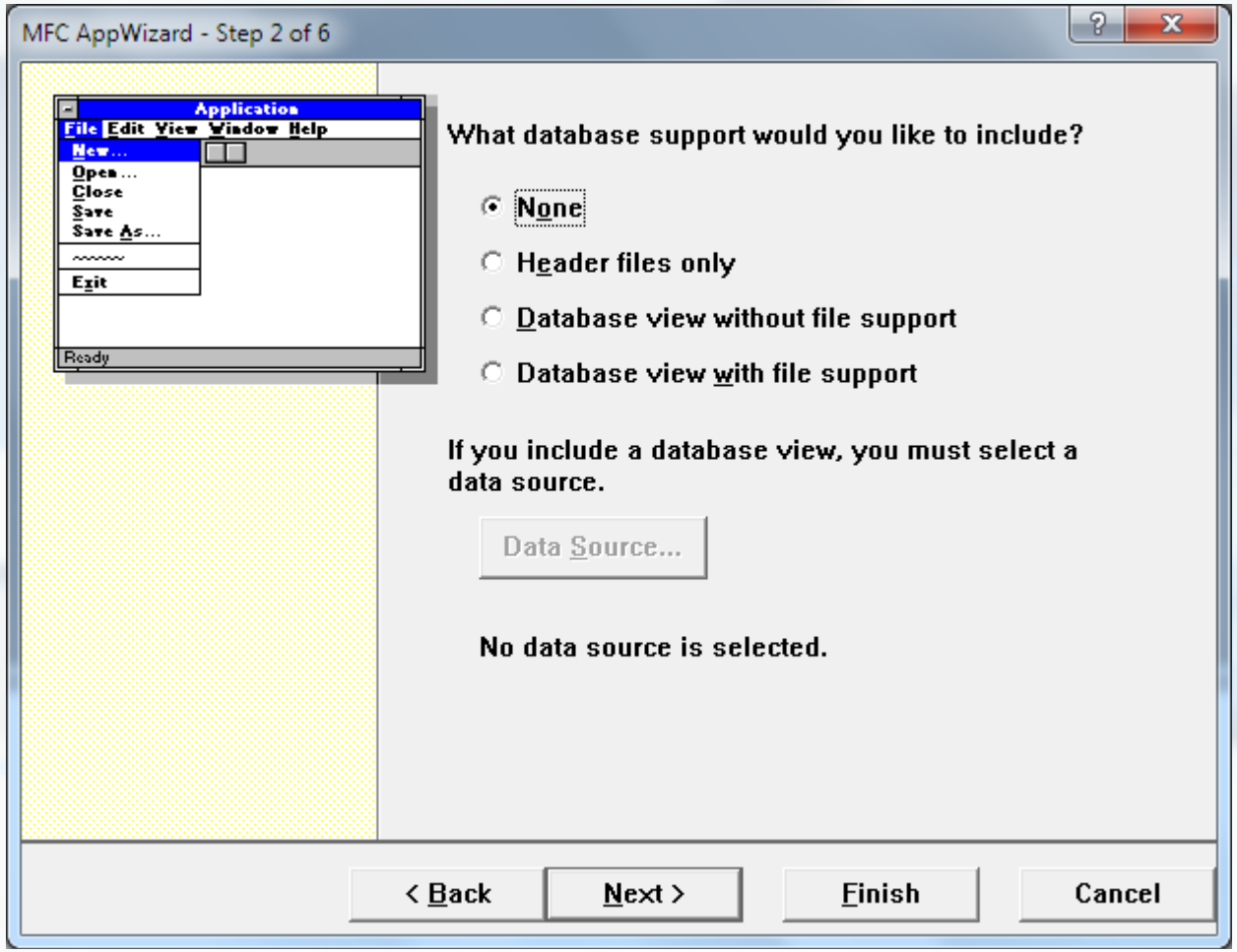
交互技术

基本过程如下：


MFC AppWizard - Step 1
Application - Document 1
File Edit View Window Help
Ready
What type of application would you like to create?
Single document
Multiple documents
Dialog based
Document/View architecture support?
What language would you like your resources in?
English [United States] (APPWZENU.DLL)
< Back
Next >
Finish
Cancel

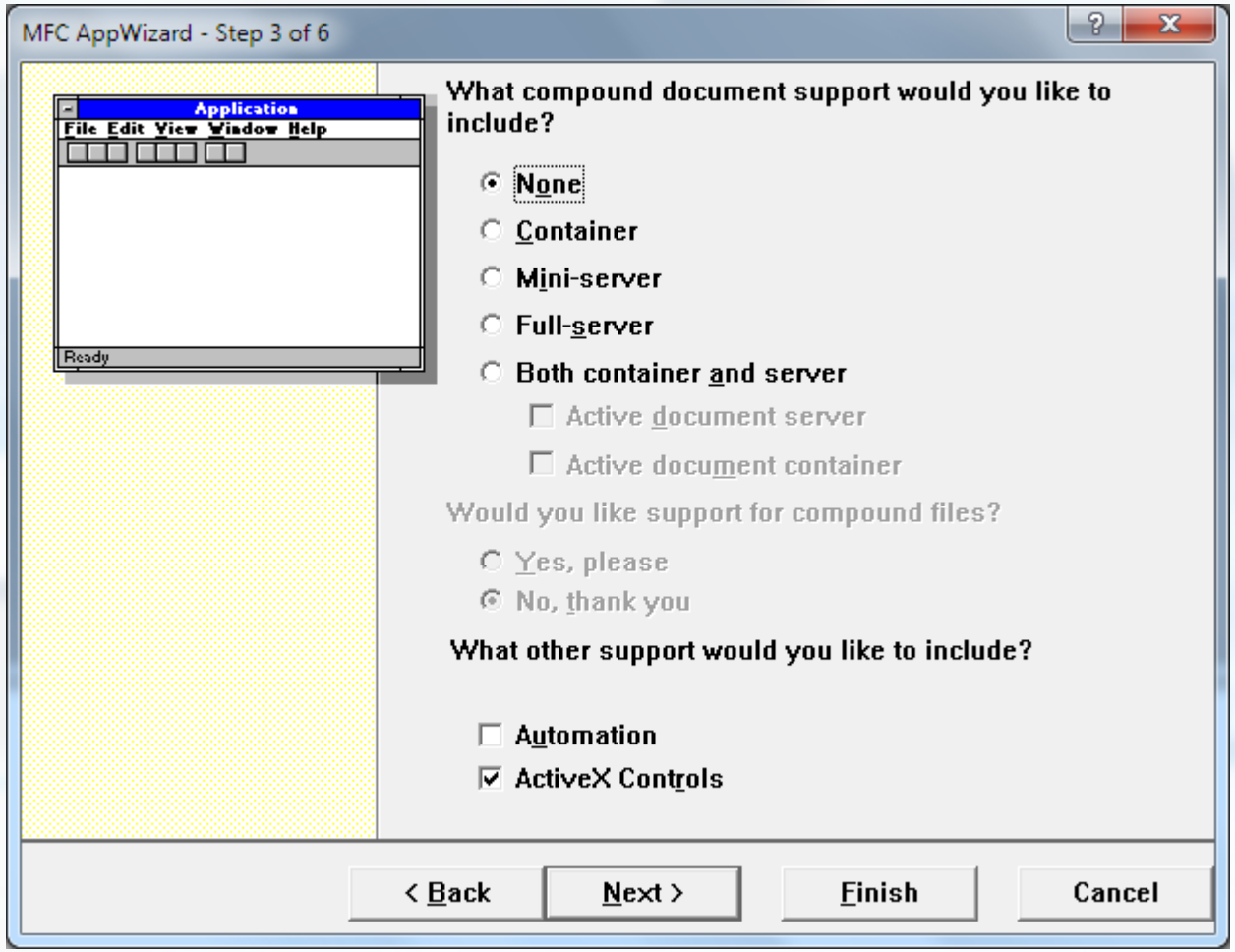
交互技术

基本过程如下：


MFC AppWizard - Step 2 of 6
Application
File Edit View Window Help
New...
Open...
Close
Save
Save As...
Exit
Ready
What database support would you like to include?
None
Header files only
Database view without file support
Database view with file support
If you include a database view, you must select a data source.
Data Source...
No data source is selected.
< Back
Next >
Finish
Cancel

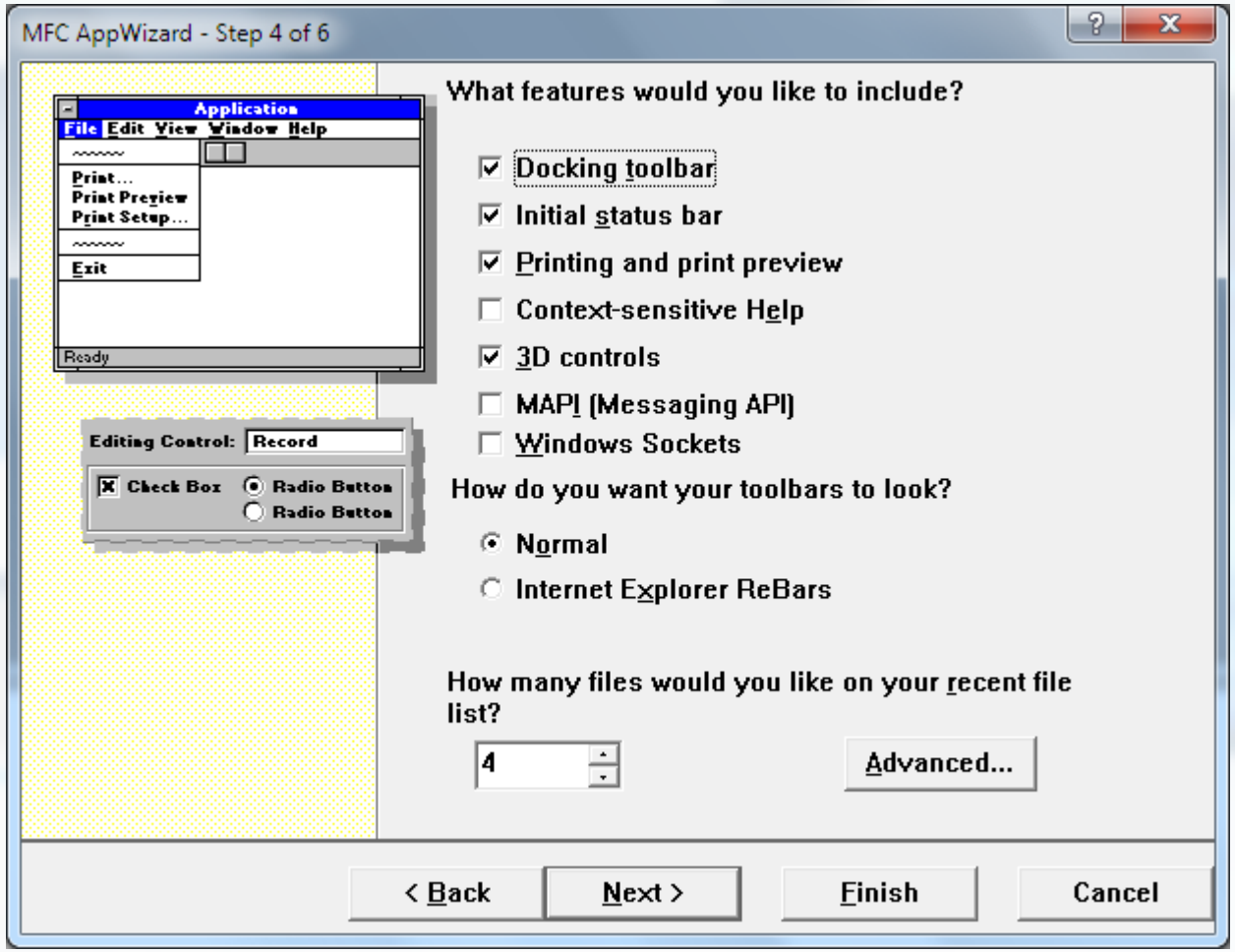
交互技术

基本过程如下：


MFC AppWizard - Step 3 of 6
Application
File Edit View Window Help
Ready
What compound document support would you like to include?
None
Container
Mini-server
Full-server
Both container and server
Active document server
Active document container
Would you like support for compound files?
Yes, please
No, thank you
What other support would you like to include?
Automation
ActiveX Controls
< Back
Next >
Finish
Cancel

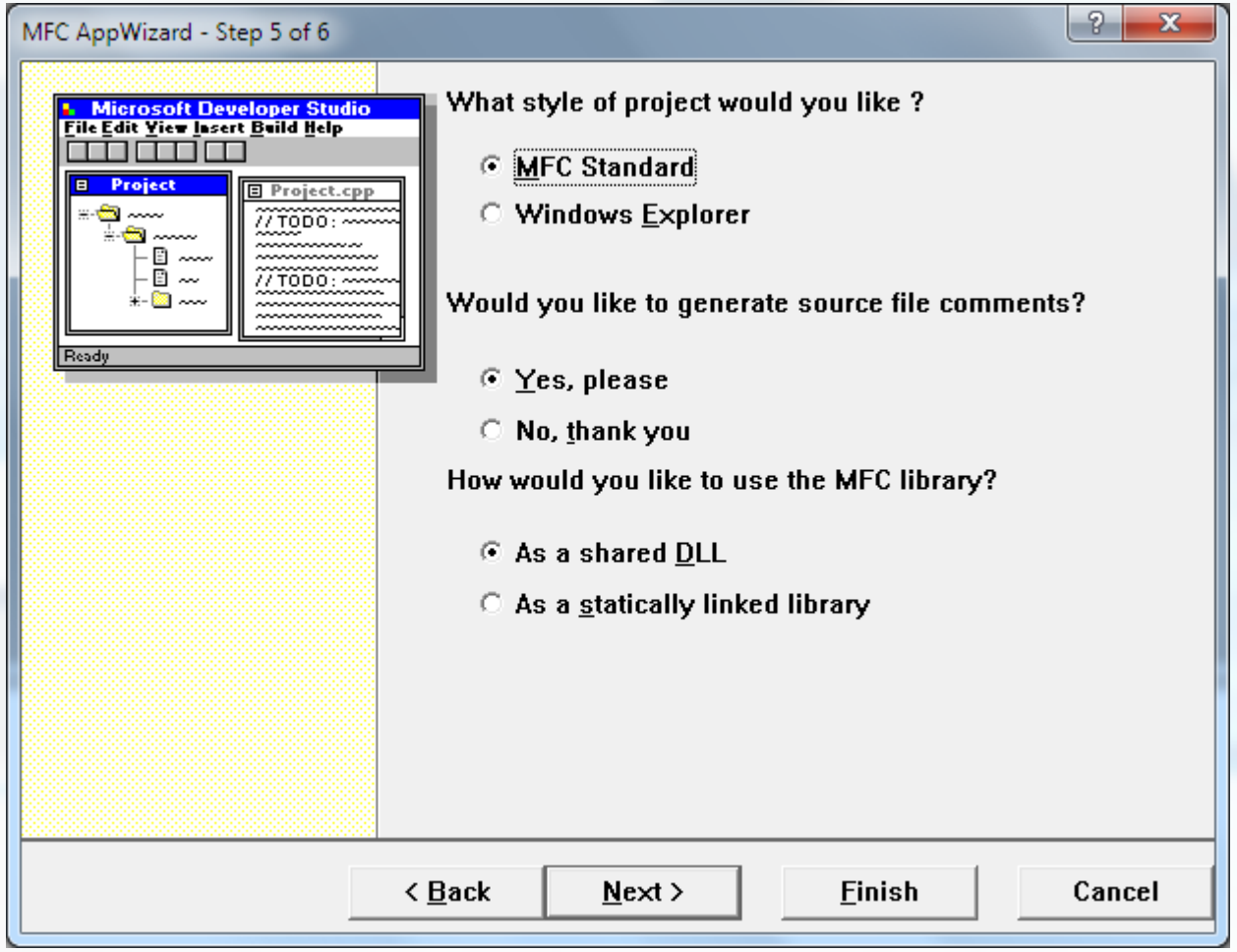
交互技术

基本过程如下：


MFC AppWizard - Step 4 of 6
Application
File Edit View Window Help
Print...
Print Preview
Print Setup...
Exit
Ready
Editing Control: Record
Check Box
Radio Button
Radio Button
What features would you like to include?
Docking toolbar
Initial status bar
Printing and print preview
Context-sensitive Help
3D controls
MAPI (Messaging API)
Windows Sockets
How do you want your toolbars to look?
Normal
Internet Explorer ReBars
How many files would you like on your recent file list?
4
Advanced...
< Back
Next >
Finish
Cancel

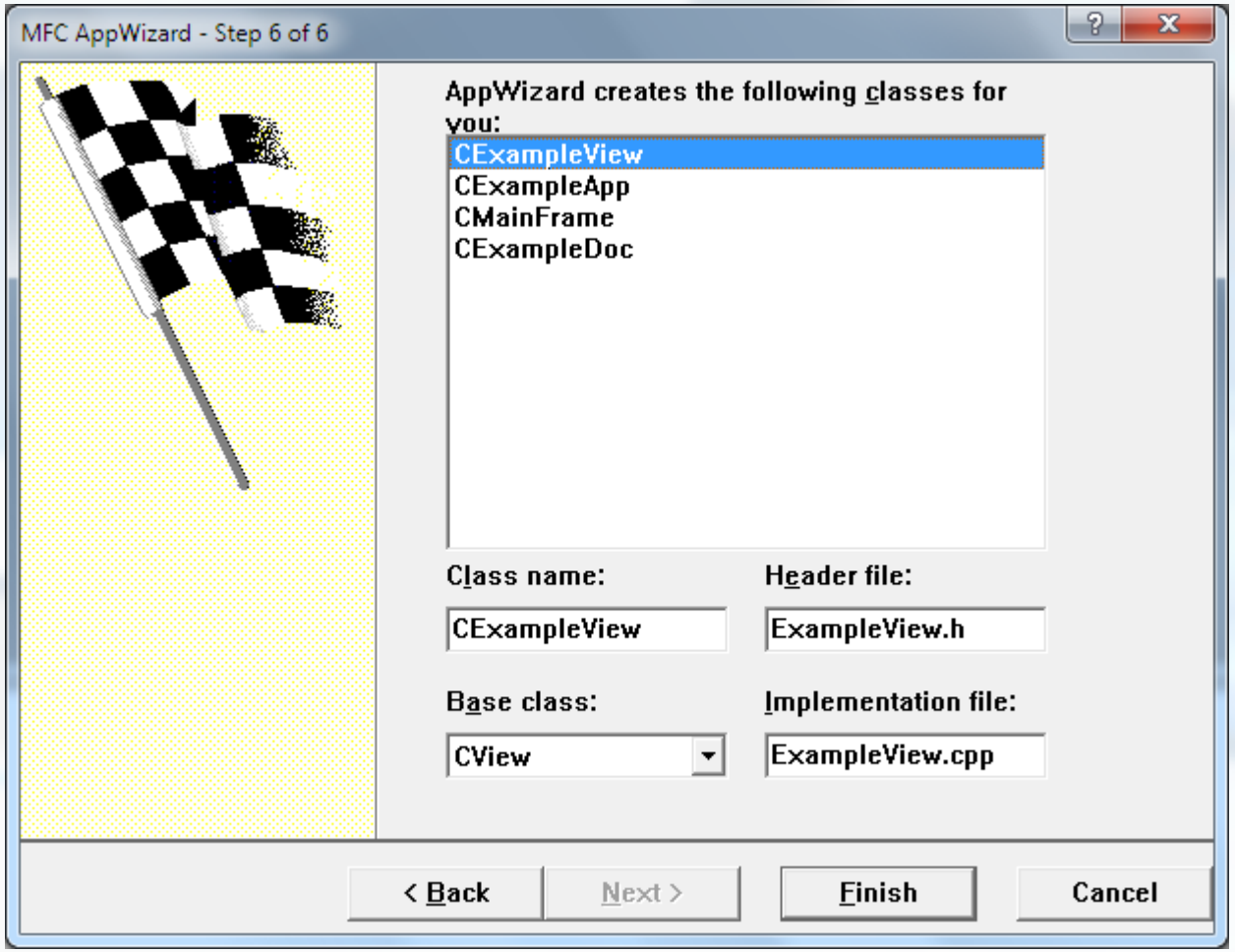
交互技术

基本过程如下：


MFC AppWizard - Step 5 of 6
Microsoft Developer Studio
File Edit View Insert Build Help
Project
Project.cpp
//TODO:
//TODO:
Ready
What style of project would you like ?
MFC Standard
Windows Explorer
Would you like to generate source file comments?
Yes, please
No, thank you
How would you like to use the MFC library?
As a shared DLL
As a statically linked library
< Back
Next >
Finish
Cancel

交互技术

基本过程如下：


MFC AppWizard - Step 6 of 6
AppWizard creates the following classes for you:
CExampleView
CExampleApp
CMainFrame
CExampleDoc
Class name:
CExampleView
Header file:
ExampleView.h
Base class:
CView
Implementation file:
ExampleView.cpp
< Back
Next >
Finish
Cancel

交互技术

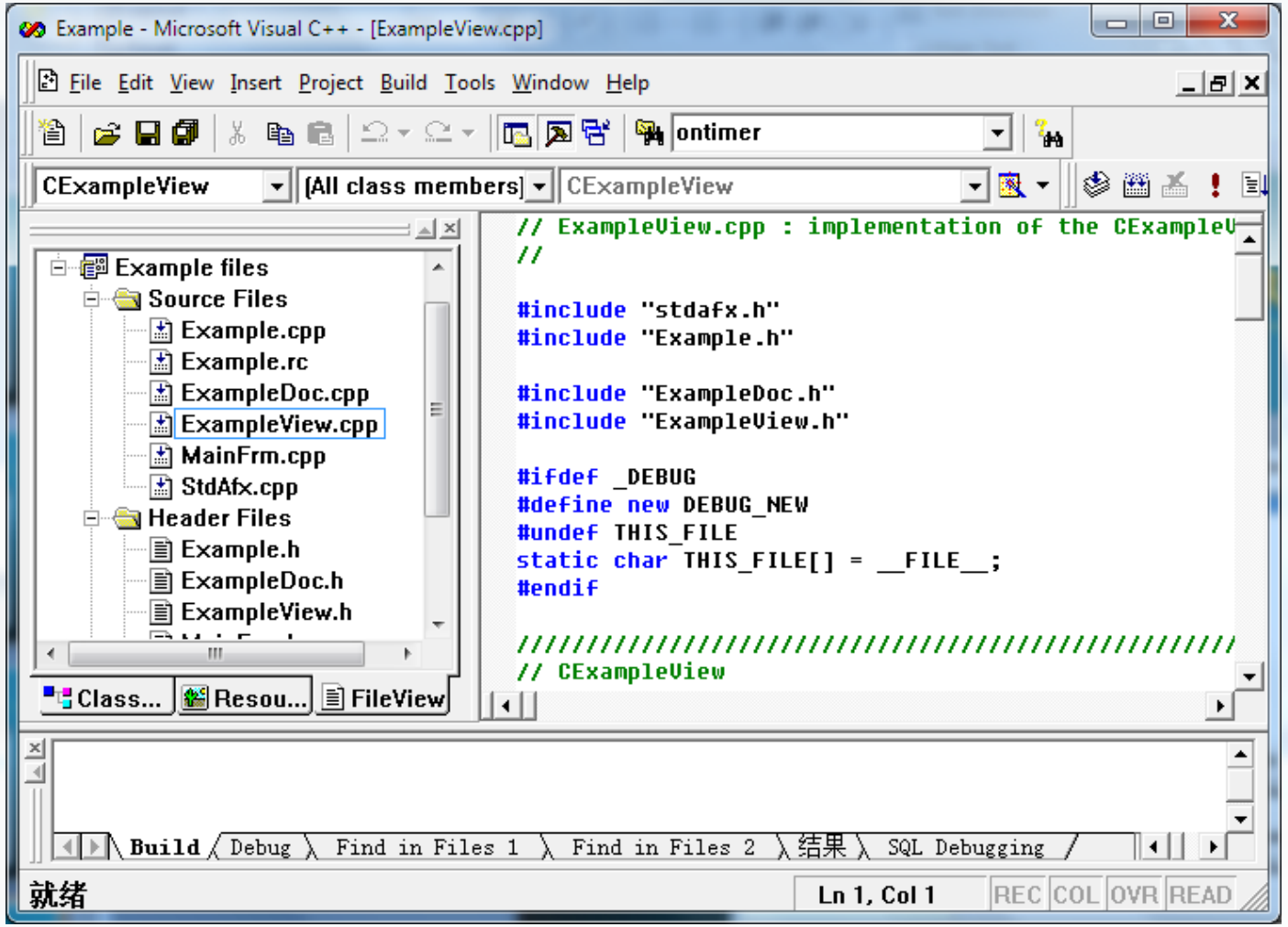
基本过程如下：

1、当用户按下鼠标左键，操作系统会向应用程序发送WM_LBUTTONDOWN消息，对应的消息函数就会被调用。

2、当用户释放鼠标左键，操作系统会向应用程序发送WM_LBUTTONUP消息，对应的消息函数就会被调用。

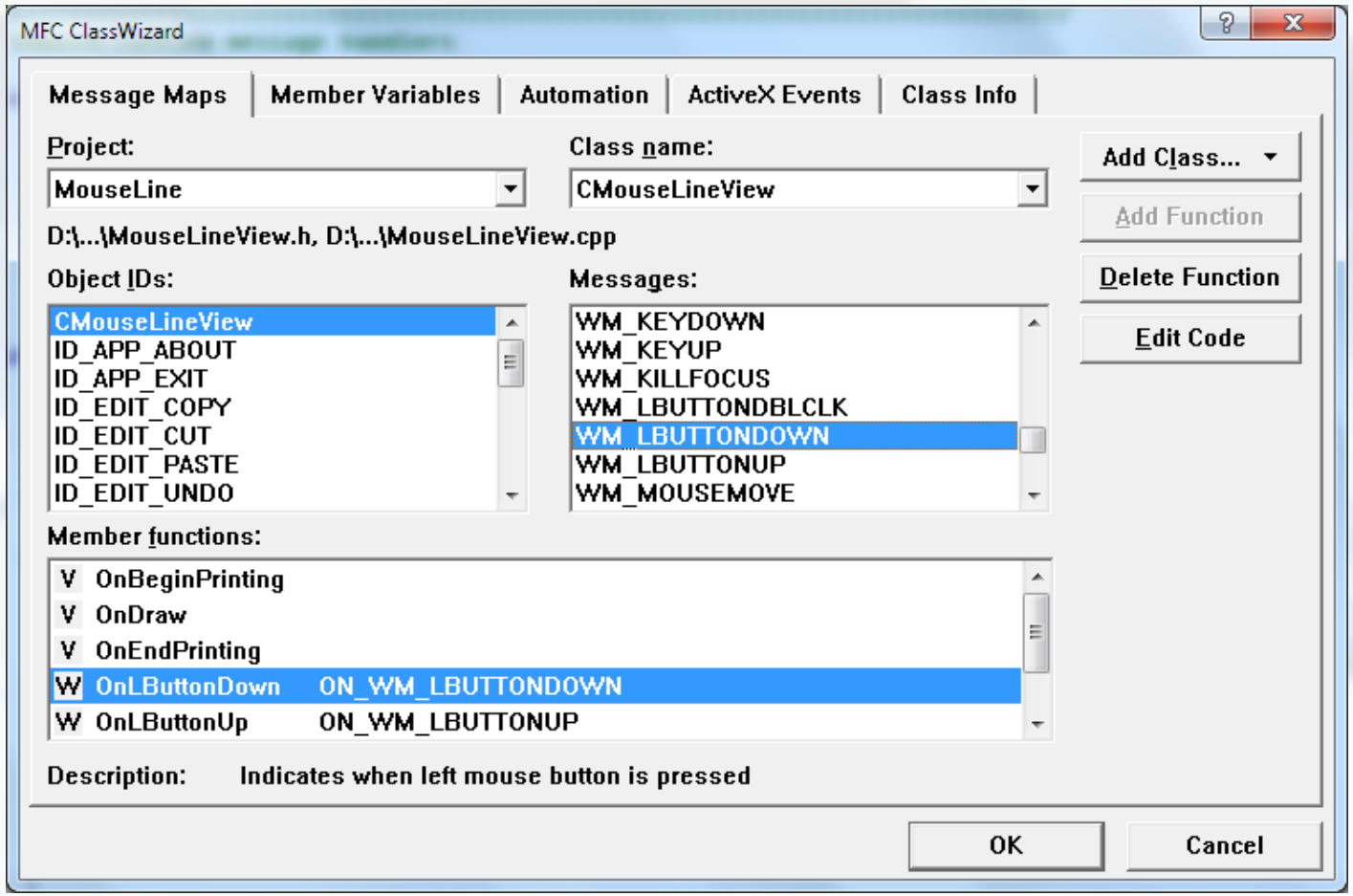
交互技术

基本过程如下：


Example - Microsoft Visual C++ - [ExampleView.cpp]
File Edit View Insert Project Build Tools Window Help
ontimer
CExampleView
(All class members)
CExampleView
Example files
Source Files
Example.cpp
Example.rc
ExampleDoc.cpp
ExampleView.cpp
MainFrm.cpp
StdAfx.cpp
Header Files
Example.h
ExampleDoc.h
ExampleView.h
Class...
Resou...
FileView
// ExampleView.cpp : implementation of the CExampleV
//
#include "stdafx.h"
#include "Example.h"
#include "ExampleDoc.h"
#include "ExampleView.h"
#ifdef _DEBUG
#define new DEBUG_NEW
#undef THIS_FILE
static char THIS_FILE[] = __FILE__;
#endif
// CExampleView
Build
Debug
Find in Files 1
Find in Files 2
结果
SQL Debugging
就绪
Ln 1, Col 1
REC COL OVR READ

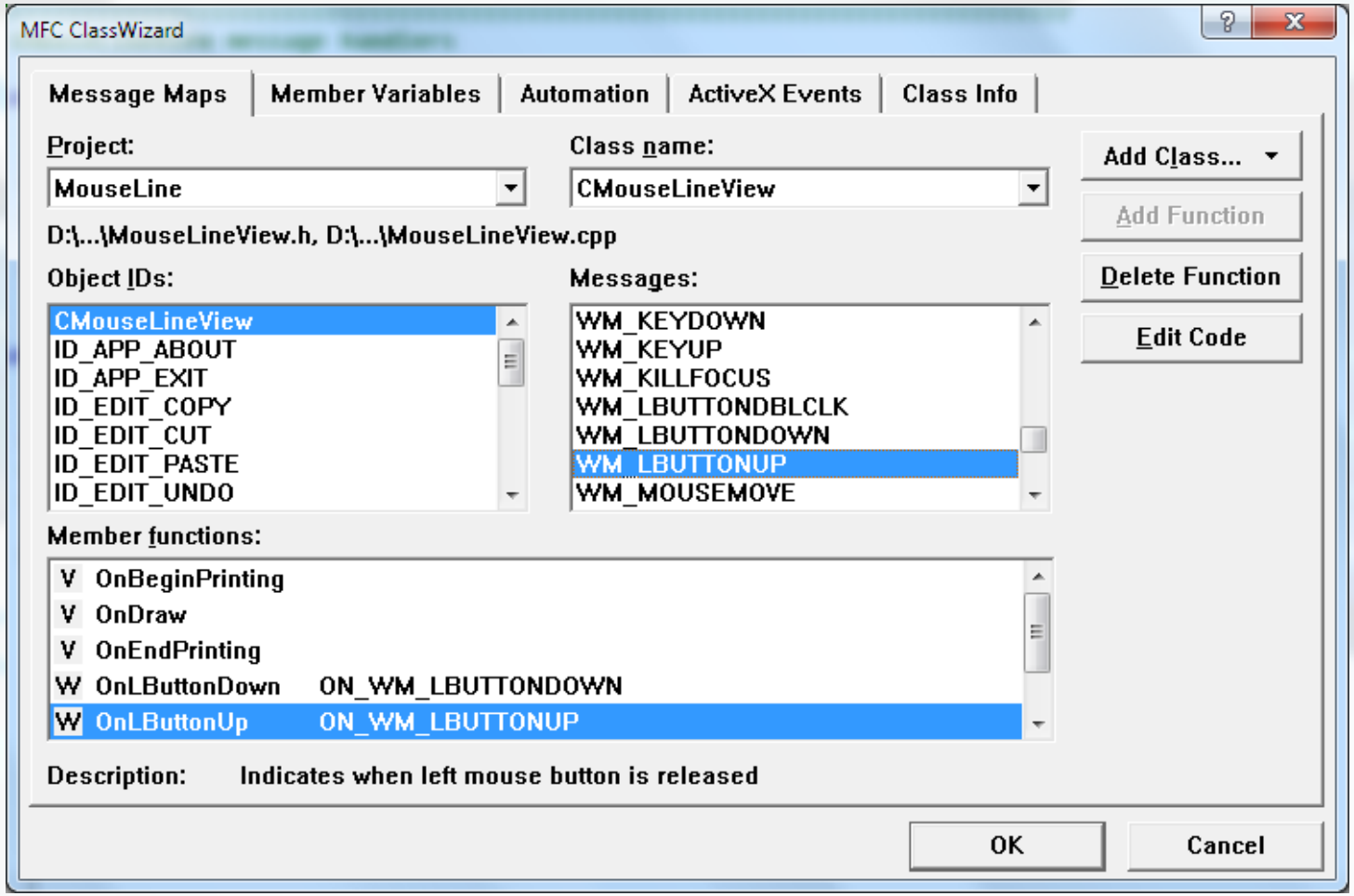
交互技术

添加WM_LBUTTONDOWN消息响应函数。



交互技术

添加WM_LBUTTONUP消息响应函数。



交互技术

用户与应用程序之间的常用交互技术包括：对话框技术、菜单技术、工具栏技术、状态栏技术。

对话框技术：可以完成用户和应用程序之间的数据交换。

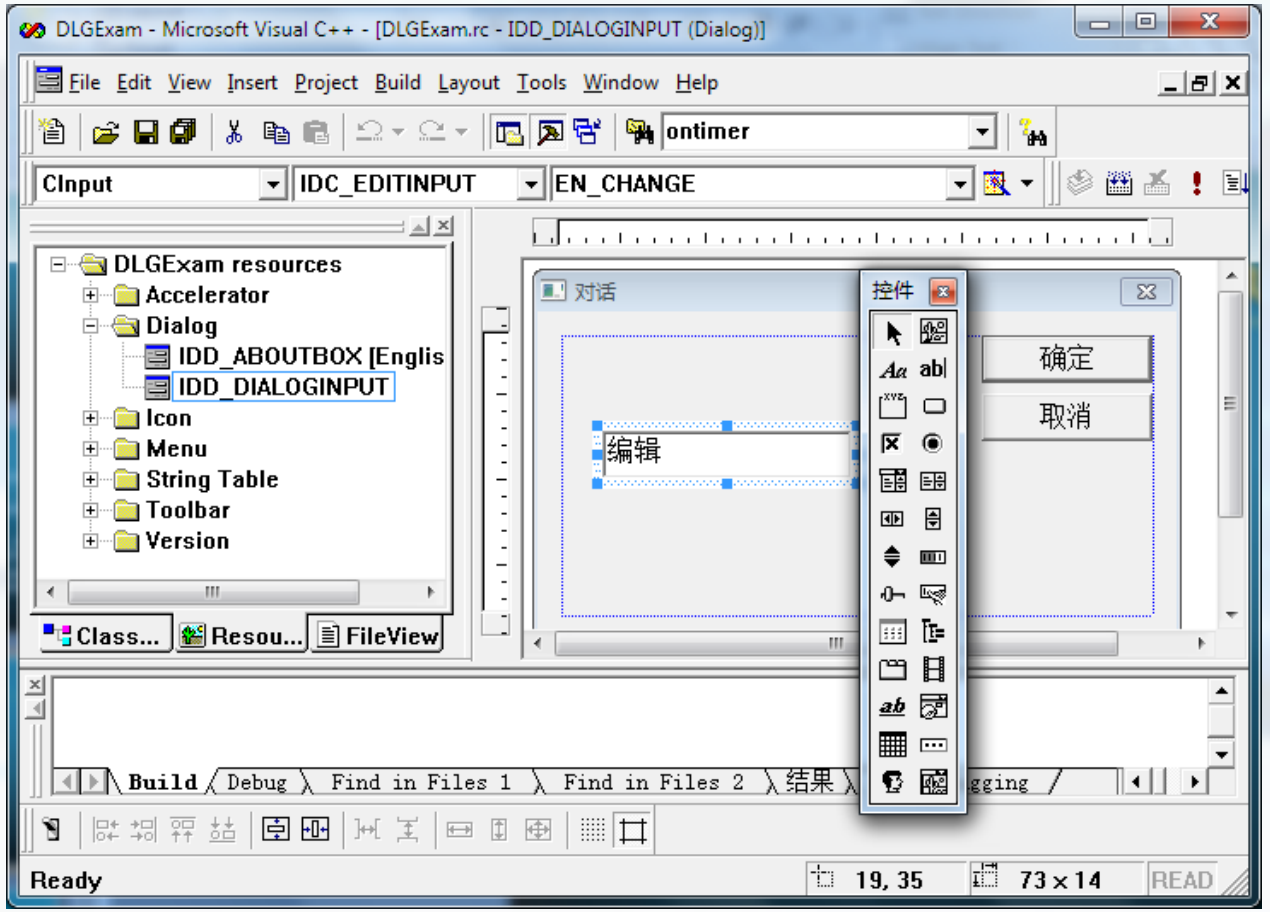
Windows下应用程序的对话框分为有模式和无模式两种。

对于有模式对话框，在对话框被关闭前，用户无法在该对话框所属的应用程序的其他地方工作。

对于无模式对话框，在该对话框显示的时候，用户仍然能将程序的焦点转向到该对话框所在的应用程序的其他地方。

交互技术

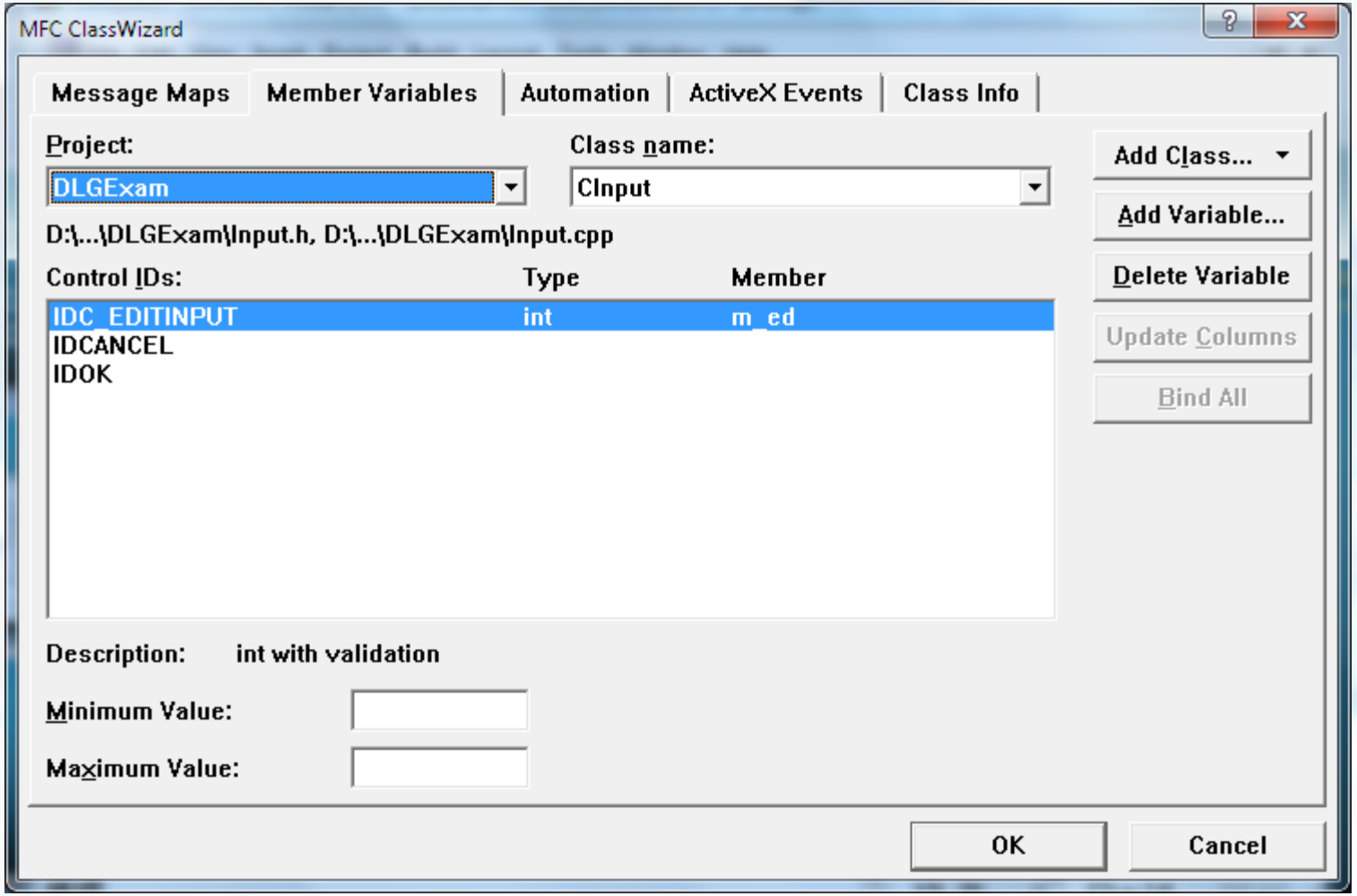
添加对话框


DLGExam - Microsoft Visual C++ - [DLGExam.rc - IDD_DIALOGINPUT (Dialog)]
File Edit View Insert Project Build Layout Tools Window Help
ontimer
CInput
IDC_EDITINPUT
EN_CHANGE
DLGExam resources
Accelerator
Dialog
IDD_ABOUTBOX [Englis
IDD_DIALOGINPUT
Icon
Menu
String Table
Toolbar
Version
Class...
Resou...
FileView
对话
编辑
确定
取消
控件
Build
Debug
Find in Files 1
Find in Files 2
结果
Ready
19, 35
73 x 14
READ

交互技术

为对话框控件

定义成员变量



交互技术

菜单技术：菜单是命令的列表，方便用户使用。

应用程序把功能类似的命令集中在一个菜单中。每一个菜单完成一个功能。设计一个菜单项时，如果是单一命令，则直接执行该命令；如果是一个较复杂的命令，则在该菜单项后加一个省略号（...），执行该命令时弹出一个对话框。

交互技术

在Win32环境下编写OpenGL程序

(1)选择菜单命令“Project-Settings...”，在“Object/library modules”处添加类库“opengl32.lib glu32.lib glaux.lib”。

(2)在视图文件的头文件中，添加以下语句

```
#include <gl\gl.h>
#include <gl\glu.h>
#include <gl\glaux.h>
```

交互技术

在视图文件的头文件变量定义中，添加以下语句

```
private:
        HGLRC hglrc;
        void init();
        void display();
```

其中， HGLRC hglrc定义一个OpenGL专用绘图描述表。对于每一个基于OpenGL的windows程序必须有专用绘图描述表RC（Rendering Context)。

交互技术

(3)绘图描述表的初始化：

在视图类中添加消息函数，分别为WM_CREATE和WM_DESTROY。其中：

```
void CDrawLineView::OnDestroy()
{
        CView::OnDestroy();
        wglDeleteContext(hglrc);
}
```

交互技术

```
int CDrawLineView::OnCreate(LPCREATESTRUCT lpCreateStruct)
{
	if (CView::OnCreate(lpCreateStruct) == -1)
		return -1;

	// TODO: Add your specialized creation code here
	PIXELFORMATDESCRIPTOR pfd={sizeof(PIXELFORMATDESCRIPTOR),
		1,
		PFD_DRAW_TO_WINDOW|PFD_SUPPORT_OPENGL|PFD_DOUBLEBUFFER,
		PFD_TYPE_RGBA,
		24,
		0,0,0,0,0,0,
		0,0,0,
		0,0,0,0,
		32,
		0,0,
		PFD_MAIN_PLANE,
		0,0,0,0};

	CClientDC clientDC(this);
	int pixelFormat = ChoosePixelFormat(clientDC.m_hDC,&pfd);
	SetPixelFormat(clientDC.m_hDC,pixelFormat,&pfd);
	hglrc = wglCreateContext(clientDC.m_hDC);
	return 0;
}
```

交互技术

(4)编写成员函数init()和display()，其中：

```
void CDrawLineView::init()
{
        glClearColor(1.0,1.0,1.0,1.0);
        glClear(GL_COLOR_BUFFER_BIT);
}
```

交互技术

```
void CDrawLineView::display()
{
        init();

        glColor3f(1.0f,0.0f,0.0f);
        glBegin(GL_LINES);
                glVertex2f(-1.0,0.0);
                glVertex2f(1.0,0.0);
                glVertex2f(0.0,-1.0);
                glVertex2f(0.0,1.0);
        glEnd();
}
```

交互技术

(5)编写成员函数OnDraw()：

```
void CDrawLineView::OnDraw(CDC* pDC)
{
        CDrawLineDoc* pDoc = GetDocument();
        ASSERT_VALID(pDoc);
        // TODO: add draw code for native data here
        wglMakeCurrent(pDC->m_hDC,hglrc);
        display();
        wglMakeCurrent(pDC->m_hDC,NULL);
        SwapBuffers(pDC->m_hDC);
}
```